

VAIPR 2.0 Exposed Limit Calc

Exposed Limit Calculation – Overview

The **Exposed Limit** calculation determines how much of the final contract/layer-adjusted **loss*** should be attributed back to each individual location.

In simple terms:

We start with **location-level gross loss**, apply **sublimit terms** and **layer terms** at grouped levels, calculate a **final contract-level loss**, and then **allocate that loss back to locations proportionally**.

***loss:** In this context, loss means maximum loss potential, not claims info

Example

This example shows how exposed limit is calculated for a selected set of locations within one contract and one peril.

Assumptions:

- Selected locations are L1, L2, L3, and L4.
- All values are already in USD.
- No facultative reinsurance is applied in this example.
- No location-level limit is binding in this example.
- The policy deductible is used as a cap on summed location deductibles within a group.
- In this example, location_gross_loss_amount is approximated as TIV minus location deductible.

Definitions:

- TIV_i = total insured value for location i
- LD_i = location deductible for location i
- GL_i = location_gross_loss_amount for location i

- Sum_Loc_GRLoss = total location gross loss across the selected locations
- SL_g = sublimit for group g
- PD = policy deductible
- PA = policy attachment point
- PL = policy limit
- PP = policy participation
- Contract_GR_Loss = final contract gross loss after sublimit and layer terms
- GR_ExpLim_i = exposed limit allocated to location i

$$S = \{L1, L2, L3, L4\}$$

Input Data

Location	TIV_i	LD_i	Sublimit
$L1$	50	10	30
$L2$	75	15	50
$L3$	100	10	30
$L4$	80	20	None

PD	50
PA	20
PL	100
PP	0.80

Where:

- (SL_g) = sublimit for group (g)
- (PA) = layer attachment point
- (PL) = layer occurrence limit
- (PP) = layer participation

Step 1: Calculate location gross loss amount

$$GL_i = \max(TIV_i - LD_i, 0)$$

For L1:

$$GL_1 = \max(50 - 10, 0) = 40$$

For L2:

$$GL_2 = \max(75 - 15, 0) = 60$$

For L3:

$$GL_3 = \max(100 - 10, 0) = 90$$

For L4:

$$GL_4 = \max(80 - 20, 0) = 60$$

So the total selected-location gross loss is:

$$Sum_Loc_GRLoss = GL_1 + GL_2 + GL_3 + GL_4 = 40 + 60 + 90 + 60 = 250$$

Step 2: Apply Sublimit Terms

$$GroupExposure_g = \max \left(\min \left(\sum_{i \in g} TIV_i - \min \left(\sum_{i \in g} LD_i, PD \right), SL_g \right), 0 \right)$$

For a group with no sublimit, the workbook logic is:

$$GroupExposure_{None} = \max \left(\sum_{i \in g} TIV_i - \min \left(\sum_{i \in g} LD_i, PD \right), 0 \right)$$

Group with sublimit 30 contains L1 and L3:

$$\begin{aligned} \sum TIV &= 50 + 100 = 150 \\ \min \left(\sum LD, PD \right) &= \min(10 + 10, 50) = 20 \\ GroupExposure_{30} &= \max(\min(150 - 20, 30), 0) = \max(\min(130, 30), 0) = 30 \end{aligned}$$

Group with sublimit 50 contains L2:

$$\begin{aligned} \sum TIV &= 75 \\ \min \left(\sum LD, PD \right) &= \min(15, 50) = 15 \\ GroupExposure_{50} &= \max(\min(75 - 15, 50), 0) = \max(\min(60, 50), 0) = 50 \end{aligned}$$

No-sublimit group contains L4:

$$\begin{aligned} \sum TIV &= 80 \\ \min \left(\sum LD, PD \right) &= \min(20, 50) = 20 \\ GroupExposure_{None} &= \max(80 - 20, 0) = 60 \end{aligned}$$

Total exposure before layer terms is:

$$PreLayer = 30 + 50 + 60 = 140$$

Step 3: Apply Layer Terms

$$Contract_GR_Loss = \min(\max(PreLayer - PA, 0), PL) \cdot PP$$

$$Contract_GR_Loss = \min(\max(140 - 20, 0), 100) \cdot 0.80$$

$$Contract_GR_Loss = \min(120, 100) \cdot 0.80 = 100 \cdot 0.80 = 80$$

So the final contract gross loss is:

$$Contract_GR_Loss = 80$$

Step 4: Back-Allocate Final Contract Gross Loss to Selected Locations

To match the selected-location output, allocate the final contract gross loss back to each location by location gross loss share:

$$GR_ExpLim_i = Contract_GR_Loss \cdot \frac{GL_i}{Sum_Loc_GRLoss}$$

For L1:

$$GR_ExpLim_1 = 80 \cdot \frac{40}{250} = 12.80$$

For L2:

$$GR_ExpLim_2 = 80 \cdot \frac{60}{250} = 19.20$$

For L3:

$$GR_ExpLim_3 = 80 \cdot \frac{90}{250} = 28.80$$

For L4:

$$GR_ExpLim_4 = 80 \cdot \frac{60}{250} = 19.20$$

Output

Location	TIV_i	GL_i	$Contract_GR_Loss$	GR_ExpLim_i
L1	50	40	80	12.80
L2	75	60	80	19.20
L3	100	90	80	28.80
L4	80	60	80	19.20

Because $Contract_GR_Loss$ is a contract-level value, the same $Contract_GR_Loss$ appears on each location row. Check:

$$12.80 + 19.20 + 28.80 + 19.20 = 80.00 = Contract_GR_Loss$$

The example calculates the contract-level exposed limit by grouping locations by sublimit, applying policy attachment, policy limit, and participation, and producing a final contract gross

loss of 80. The selected-location output then back-allocates that Contract_GR_Loss to each location in proportion to the location's share of total location gross loss.

i Inputs:

- For each selected location:
 - Replacement values (RV1, RV2, RV3, RV4)
 - Total replacement value (TRV)
 - Location deductible type and deductible amounts
 - Location limit type and limit amounts
 - Location participation values
- Sublimit inputs:
 - Sublimit amount
 - Sublimit attachment
 - Sublimit minimum and maximum deductible rules
- Layer inputs:
 - Layer attachment point
 - Layer occurrence limit
 - Layer participation
 - Layer deductible rules

...

i Output

- contract_number
- insured_name
- address_text
- city
- area_code
- postal_code
- country_2_digit_iso_code
- country_name
- cede_db
- peril
- original_currency_code
- exchange_rate_used
- TIV
- location_gross_loss_amount
- Sum_Loc_GRLoss
- location_deductible_amount
- location_interim_limit_amount
- sublimit_attachment_point
- layer_occupation_total_limit_amount
- layer_occupation_participation
- layer_attachment_point
- Contract_GR_Loss
- GR_ExpLim

High-Level Flow

The calculation happens in **4 main stages**:

1. **Filter the data to the relevant peril**
 2. **Apply sublimit terms**
 3. **Apply layer terms**
 4. **Allocate the final adjusted loss back to each location**
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Step 1: Filter to the relevant exposure records

The process begins by selecting exposure records for the required peril:

- E.g., records where `peril = 'FF'`
- Data comes from:
 - `vaipr_location`
 - `vaipr_location_exposure`

This creates the base dataset used throughout the rest of the calculation.

Step 2: Apply Sublimit Terms

The code then groups losses at the **sublimit correlation/group level** using:

- `contract_key`
- `layer_sid`
- `limit_correlation_id`
- `cede_db`

At this grouped level, the following are calculated:

- Total location deductible
- Total gross loss
- Sublimit amount
- Minimum/maximum deductible constraints
- Sublimit attachment point

What happens in this stage?

2.1 Determine the sublimit deductible

The deductible used for the group is adjusted based on:

- **minimum deductible**
- **maximum deductible**
- whether the deductible is stored as:
 - a flat amount, or
 - a percentage of gross loss

2.2 Apply the sublimit attachment

If an attachment exists:

- Loss below the attachment is reduced to zero
- Loss above the attachment is reduced by the attachment amount

2.3 Adjust for deductible differences

Because some deductible has already been applied at location level, the code adjusts the grouped loss so that the final deductible at sublimit level is consistent.

2.4 Apply the sublimit cap

If the adjusted grouped loss exceeds the sublimit, it is capped at the sublimit amount.

Business meaning

At the end of this stage, each sublimit group has a **post-sublimit adjusted loss**.

Step 3: Apply Layer Terms

After sublimit processing, the code rolls the results up to the **layer level**.

This uses:

- `contract_key`
- `layer_sid`
- `cede_db`

It also joins in the relevant **layer configuration**, including:

- layer deductible type
- attachment point
- occurrence limit

- participation
- deductible values

What happens in this stage?

3.1 Calculate the layer deductible

The deductible applied depends on `layer_deductible_type_code` .

The code supports different deductible styles, such as:

- flat deductible
- minimum deductible
- maximum deductible
- percentage-of-loss deductible
- franchise/attachment-based handling

3.2 Adjust the loss for the layer deductible

If the layer deductible differs from what was already applied during sublimit processing, the grouped loss is adjusted accordingly.

3.3 Apply the layer attachment point

If a layer attachment exists:

- losses below the attachment are removed
- losses above the attachment are reduced by the attachment amount

3.4 Apply occurrence limit and participation

The adjusted layer loss is then subject to:

- **occurrence total limit**
- **participation percentage**

This means:

- if the loss exceeds the occurrence limit, it is capped
- then only the participating share is retained

Business meaning

At the end of this stage, the model has a **final layer-adjusted loss**.

Step 4: Roll up to Contract Level

Once all layer terms have been applied, the code sums the final layer-adjusted losses by `contract_key`.

Business meaning

This is the total loss remaining for the contract after all sublimit and layer rules have been applied.

Step 5: Allocate the Final Contract Loss Back to Locations

This is the final step where **Exposed Limit** is created.

The code:

1. Calculates the total original location gross loss for the contract.
2. Allocates the final contract loss back to each location based on that location's share of the original gross loss

Final allocation formula

$$GR_ExpLim = Contract\ Final\ Loss \times \frac{Location\ Gross\ Loss}{Total\ Contract\ Gross\ Loss}$$

Business meaning

Each location receives a proportion of the final contract loss based on how much of the original contract gross loss it contributed.

Exposed Limit – Summary

The **Exposed Limit** is calculated by first applying all relevant **sublimit** and **layer** terms to the gross loss for each peril, and then allocating the resulting **final contract-level loss** back to each location.

High-level steps

1. Filter exposure records to a required peril
2. Group losses by sublimit group and apply:
 - deductible rules
 - attachment
 - sublimit cap
3. Group results to layer level and apply:

- layer deductible
- layer attachment
- occurrence limit
- participation

4. Sum the adjusted loss to contract level

5. Allocate the final contract loss back to each location based on its share of the original gross loss

Final allocation formula

$$\textit{Exposed Limit per Location} = \textit{Final Contract Loss} \times \frac{\textit{Location Gross Loss}}{\textit{Total Contract Gross Loss}}$$